Citation Sources

Bibliographic and provenance record

Abstraction Mechanisms for Virtual Avatars

Nidanur Günay · University of Konstanz · 2026

What this document is. One record per source cited in the thesis, ordered by first appearance. Each gives the verified bibliographic fields, the DOI or URL, the path to the local copy, and every place in the thesis where the source is cited.
What it is not. It is not the verification checklist. Sentence-by-sentence checking of every citation against its source lives in `citation_check.pdf`, which carries one tickable entry per citation instance. This document answers *what is this source and where did it come from*; that one answers *does this sentence match it*.
Reading the status line. ✓ marks a source read at primary-source depth during the citation audit of 21 August 2026, where the summary states what the paper actually contains. ○ marks a source whose bibliographic fields are verified but whose content summary has not been written; nothing is asserted about its contents here.

1 weidner2023

Weidner, Florian, Boettcher, Gerd, Arevalo Arboleda, Stephanie, Diao, Chenyao, Sinani, Luljeta, Kunert, Christian, Gerhardt, Christoph, Broll, Wolfgang, Raake, Alexander. *A Systematic Review on the Visualization of Avatars and Agents in AR & VR Displayed Using Head-Mounted Displays*. IEEE Transactions on Visualization and Computer Graphics. vol. 29, no. 5, pp. 2596–2606. 2023.

DOI / URL	`https://doi.org/10.1109/TVCG.2023.3247072`
Local copy	`02_trust_and_avatar_perception/Weidner2023.pdf`
Cited in	Ch.1 Introduction → Motivation (1 citation)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

2 mcdonnell2012

McDonnell, Rachel, Breidt, Martin, Bülthoff, Heinrich H.. *Render Me Real? Investigating the Effect of Render Style on the Perception of Animated Virtual Humans*. ACM Transactions on Graphics. vol. 31, no. 4, art. 91, pp. 91:1–91:11. ACM. 2012.

DOI / URL	`https://doi.org/10.1145/2185520.2185587`
Local copy	`01_uncanny_valley/ McDonnell, R., Breidt, M., and BÃijlthoff, H. H. (2012) . Render me Real? Investigating.pdf`
Cited in	Ch.1 Introduction → Motivation Ch.3 Related Work → Research Gap Ch.3 Related Work → The Uncanny Valley and Avatar Realism Ch.4 Methodology → NPR Technique Design Ch.4 Methodology → User Study Design (5 citations)

What it is. Ten render styles spanning photorealistic to fully abstract, applied to identical geometry and motion, rated on realism, appeal, re-assurance, friendliness and trustworthiness, for still images and video separately.

What it establishes. A valley-shaped pattern: four of the most abstract styles and the two most realistic ones scored highest on appeal, re-assurance and trustworthiness alike. But the authors hedge twice. They "cannot say that all abstract cartoon characters are considered appealing" and name ToonBare as their counter-example, and they report "our most realistic character was often rated as appealing or pleasant as our cartoon characters", which they call "contrary to the theory". Their practitioner take-away is a reassurance about realism, not an endorsement of stylisation.

✓ Read at primary-source depth during the 2026-08-21 audit.

3 natarajan2020

Natarajan, Manisha, Gombolay, Matthew. *Effects of Anthropomorphism and Accountability on Trust in Human Robot Interaction*. In: Proceedings of the 2020 ACM/IEEE International Conference on Human-Robot Interaction. pp. 33–42. 2020.

DOI / URL `https://doi.org/10.1145/3319502.3374839`

Local copy `02_trust_and_avatar_perception/Natarajan-Effects of Anthropomorphism and Accountability.pdf`

Cited in Ch.1 Introduction → Motivation (1 citation)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

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4 mori1970

Mori, Masahiro. *The Uncanny Valley*. Energy. vol. 7, no. 4, pp. 33–35. 1970.

DOI / URL *none recorded*

Local copy `01_uncanny_valley/The Uncanny Valley- The Original Essay by Masahiro Mori.pdf`

Cited in Ch.1 Introduction → Motivation Ch.3 Related Work → The Uncanny Valley and Avatar Realism Ch.3 Related Work → Trust and Credibility in Virtual Agents Ch.4 Methodology → Research Questions Ch.4 Methodology → User Study Design (5 citations)

What it is. The founding essay. Mori proposes that affinity (*shinwakan*) rises with human likeness up to a point, drops sharply for entities that are almost but not quite human, then recovers near full likeness. He plots separate curves for still and moving figures, the moving curve descending far deeper. His reference points inside the valley are a prosthetic hand, a corpse and the living dead.

What it establishes. Crucially, the eerie response is *delayed*: "we cannot distinguish at a glance a prosthetic hand from a real one" (PDF p. 2), and the sensation arrives only "when we realize the hand. . . is in fact artificial" or during a handshake (p. 3). He also names several flaws, not one. A myoelectric hand that moves "could make healthy people feel uneasy" (p. 3), which is his own motion example.

✓ Read at primary-source depth during the 2026-08-21 audit.

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5 canales2024

Canales, Ryan, Roble, Doug, Neff, Michael. *The Impact of Avatar Stylization on Trust*. In: 2024 IEEE Conference on Virtual Reality and 3D User Interfaces (VR). pp. 418–428. 2024.

DOI / URL `https://doi.org/10.1109/VR58804.2024.00063`

Local copy `02_trust_and_avatar_perception/Avatar_Stylization_and_Trust_IEEE_VR_2024.pdf`

Cited in Ch.1 Introduction → Motivation Ch.1 Introduction → Problem Statement Ch.3 Related Work → NPR for Virtual Humans Ch.3 Related Work → Research Gap Ch.3 Related Work → Trust and Credibility in Virtual Agents Ch.4 Methodology → User Study Design Ch.6 Evaluation → User Study Results (7 citations)

What it is. VR study, $n = 66$. Participants imagine a possible cancer diagnosis and interview three doctors, each a different identity (AF, AM, CM) at a different stylisation level (Real, Mid, Caricature), so identity is crossed with stylisation. Stylisation varied sculpted geometry and authored textures; all levels rendered through the same Unreal default material.

What it establishes. Measured after *each* doctor: Ability, Benevolence, Integrity and a direct Trust block, 7-point Likert; propensity to trust once at the start; a forced doctor choice at the end. Linear mixed-effects models. "Style generally had no impact on ratings, with one partial exception for benevolence" ($p = .034$, post-hocs unresolved, Real trending lowest). Behaviourally the Mid style was chosen 48 % of the time versus 26 % each for the others.

✓ Read at primary-source depth during the 2026-08-21 audit.

6 rahman2024

Rahman, Md. Shahinur, Sabbir, Md. Mahiuddin, Zhang, Jing, Taufique, Khan Md. Raziuddin, Rahman, Md. Masudur. *Navigating Avatar-Based Marketing From Value Aspects: A Systematic Literature Review and Forward-Looking Research Agenda*. International Journal of Consumer Studies. vol. 48, no. 6, pp. e13096. 2024.

DOI / URL `https://doi.org/10.1111/ijcs.13096`
Local copy `02_trust_and_avatar_perception/Navigating_Avatar-Based_Marketing_From_Value_Aspec.pdf`
Cited in Ch.1 Introduction → Motivation (1 citation)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

7 isenberg2003

Isenberg, Tobias, Freudenberg, Bert, Halper, Nick, Schlechtweg, Stefan, Strothotte, Thomas. *A Developer's Guide to Silhouette Algorithms for Polygonal Models*. IEEE Computer Graphics and Applications. vol. 23, no. 4, pp. 28–37. 2003.

DOI / URL `https://doi.org/10.1109/MCG.2003.1210862`
Local copy `03_npr_rendering/Isenberg_2003_ADG.pdf`
Cited in Ch.1 Introduction → Problem Statement Ch.3 Related Work → NPR for Virtual Humans Ch.4 Methodology → NPR Technique Design (3 citations)

What it is. Practitioner survey ("Survey" banner on p. 1) guiding developers in choosing a silhouette algorithm. Explicitly restricted to polygonal models.

What it establishes. Organising idea: every silhouette algorithm must solve detection and then visibility culling. Classes: image-space, hybrid, object-space. Recommends image-space and hybrid for real-time work; precomputation methods "make animation of the models inefficient". Also fixes the edge vocabulary: the *contour* is the subset of the silhouette separating object from background, *internal silhouettes* lie inside it, and creases, borders and self-intersections "together with the silhouette lines are often called feature lines". Back-face expansion is documented as a two-pass hybrid technique. The word *abstraction* appears once in the whole article, inside a reference title.

✓ Read at primary-source depth during the 2026-08-21 audit.

8 wallraven2007

Wallraven, Christian, Bülthoff, Heinrich H., Cunningham, Douglas W., Fischer, Jan, Bartz, Dirk. *Evaluation of Real-World and Computer-Generated Stylized Facial Expressions*. ACM Transactions on Applied Perception. vol. 4, no. 3, art. 16, pp. 16:1–16:24. ACM. 2007.

DOI / URL `https://doi.org/10.1145/1278387.1278390`
Local copy `Wallraven_Stylization_TAP2007.pdf`
Cited in Ch.1 Introduction → Problem Statement Ch.4 Methodology → NPR Technique Design (2 citations)

What it is. Three stylisation techniques (brush-stroke, cartoon, illustrative) applied at three abstraction levels, to both a CG avatar (Experiment 2) and real-world video of actors (Experiment 3). Measures: recognition, response time, intensity, sincerity, preference.

What it establishes. The source for abstraction-versus-intensity: "perceived intensity is dependent on the resolution level. . . **higher abstraction levels result in less intensity**" (PDF p. 21). On recognition against real footage: "both the original video sequences as well as the cartoon stylization were equally well recognized, whereas there was a significant drop in performance for the brush and illustrative methods" (p. 18). Cartoon stylisation of real video was statistically indistinguishable from the original on recognition, speed, intensity and sincerity.

✓ Read at primary-source depth during the 2026-08-21 audit.

9 unityurp

Unity Technologies. *Inject a Render Pass with a Scriptable Renderer Feature in URP*. 2026.

DOI / URL	*none recorded*
Local copy	`https://docs.unity3d.com/6000.0/Documentation/Manual/urp/renderer-features/scriptable-renderer-features/inject-a-pass-using-a-scriptable-renderer-feature.html`
Cited in	Ch.2 Background → Rendering Concepts (1 citation)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

10 luebke2003

Luebke, David, Reddy, Martin, Cohen, Jonathan D., Varshney, Amitabh, Watson, Benjamin, Huebner, Robert. *Level of Detail for 3D Graphics*. Morgan Kaufmann. 2003.

DOI / URL	ISBN 978-1-55860-838-2
Local copy	`04_graphics_foundations/Level of Detail for 3D Graphics-Luebke.pdf`
Cited in	Ch.2 Background → Rendering Concepts (1 citation)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

11 lewis2014

Lewis, John P., Anjyo, Ken, Rhee, Taehyun, Zhang, Mengjie, Pighin, Frédéric H., Deng, Zhigang. *Practice and Theory of Blendshape Facial Models*. In: Eurographics 2014 — State of the Art Reports. pp. 199–218. 2014.

DOI / URL	`https://doi.org/10.2312/egst.20141042`
Local copy	`04_graphics_foundations/Lewis - 2014-EG-blendshape_STAR.pdf`
Cited in	Ch.2 Background → Rendering Concepts (1 citation)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

12 nowak2005

Nowak, Kristine L., Rauh, Christian. *The Influence of the Avatar on Online Perceptions of Anthropomorphism, Androgyny, Credibility, Homophily, and Attraction*. Journal of Computer-Mediated Communication. vol. 11, no. 1, pp. 153–178. 2005.

DOI / URL	`https://doi.org/10.1111/j.1083-6101.2006.tb00308.x`
Local copy	`02_trust_and_avatar_perception/Nowak Rauh 2005.pdf`
Cited in	Ch.2 Background → Avatars, Trust and Eeriness Ch.3 Related Work → Trust and Credibility in Virtual Agents (2 citations)

What it is. 255 participants rate a series of static avatar images on credibility and related dimensions. Both anthropomorphism and gender expression predicted credibility ratings.

What it establishes. The paper never ranks appearance against other cues. Its own framing is hedged and conditional on mediation: “When people interact through computer-mediated channels, physical information

about the other person may be unavailable…people…use avatars, *or whatever information the interface provides*". The appearance-transfer claim is itself attributed to Rauh, Polonsky & Buck (2004).

✓ Read at primary-source depth during the 2026-08-21 audit.

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13 mayer1995

Mayer, Roger C., Davis, James H., Schoorman, F. David. *An Integrative Model of Organizational Trust*. Academy of Management Review. vol. 20, no. 3, pp. 709–734. 1995.

DOI / URL	`https://doi.org/10.5465/amr.1995.9508080335`
Local copy	`02_trust_and_avatar_perception/Mayer-IntegrativeModelOrganizational-1995.pdf`
Cited in	Ch.2 Background → Avatars, Trust and Eeriness Ch.3 Related Work → Research Gap Ch.3 Related Work → Trust and Credibility in Virtual Agents Ch.4 Methodology → User Study Design (4 citations)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

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14 mcknight2001

McKnight, D. Harrison, Chervany, Norman L.. *What Trust Means in E-Commerce Customer Relationships: An Interdisciplinary Conceptual Typology*. International Journal of Electronic Commerce. vol. 6, no. 2, pp. 35–59. 2001.

DOI / URL	`https://doi.org/10.1080/10864415.2001.11044235`
Local copy	`02_trust_and_avatar_perception/What Trust Means in E-Commerce Customer Relationships.pdf`
Cited in	Ch.2 Background → Avatars, Trust and Eeriness Ch.3 Related Work → Research Gap Ch.3 Related Work → Trust and Credibility in Virtual Agents Ch.4 Methodology → User Study Design (4 citations)

What it is. Conceptual paper, not an experiment. Surveys roughly eighty works on trust, extracts 65 usable definitions and sorts them into an interdisciplinary typology so results can be compared across fields. Setting is e-commerce customer–vendor relationships.

What it establishes. Four conceptual constructs: Disposition to Trust (psychology), Institution-based Trust (sociology), Trusting Beliefs and Trusting Intentions (social psychology). **Trusting Beliefs includes Competence, Benevolence, Integrity and Predictability.** Predictability enters from the economics literature and they position it *against* Mayer: "This differs from Mayer et al. [45], who excluded predictability from their trust typology." The trustee throughout is "another person, institution, or people generally" — never a machine. Across the 14 pages read: `human-computer` 0 hits, `HCI` 0 hits.

✓ Read at primary-source depth during the 2026-08-21 audit.

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15 bartneck2009

Bartneck, Christoph, Kulić, Dana, Croft, Elizabeth, Zoghbi, Sonja. *Measurement Instruments for the Anthropomorphism, Animacy, Likeability, Perceived Intelligence, and Perceived Safety of Robots*. International Journal of Social Robotics. vol. 1, no. 1, pp. 71–81. 2009.

DOI / URL	`https://doi.org/10.1007/s12369-008-0001-3`
Local copy	`02_trust_and_avatar_perception/Measurement Instruments for the Anthropomorphism, Animac pdf`
Cited in	Ch.2 Background → Avatars, Trust and Eeriness Ch.3 Related Work → Research Gap Ch.3 Related Work → Trust and Credibility in Virtual Agents Ch.4 Methodology → User Study Design Ch.4 Methodology → User Study Overview (5 citations)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

16 ho2010

Ho, Chin-Chang, MacDorman, Karl F.. *Revisiting the Uncanny Valley Theory: Developing and Validating an Alternative to the Godspeed Indices*. Computers in Human Behavior. vol. 26, no. 6, pp. 1508–1518. 2010.

DOI / URL	`https://doi.org/10.1016/j.chb.2010.05.015`
Local copy	`02_trust_and_avatar_perception/Ho2010UncannyValleyIndices.pdf`
Cited in	Ch.2 Background → Avatars, Trust and Eeriness Ch.3 Related Work → Research Gap Ch.3 Related Work → Trust and Credibility in Virtual Agents Ch.4 Methodology → User Study Design Ch.4 Methodology → User Study Overview (5 citations)

What it is. Develops and validates an alternative to the Godspeed indices for uncanny valley research, using video clips of animated characters and humanoid robots across two rounds of testing.

What it establishes. The final validated set is **19 semantic differential items across three indices**: humanness (6 items), eeriness (8) and attractiveness (5). A warmth index was also constructed during development for decorrelation testing. The key result for avatar work: the eeriness index has "no significant correlation with the humanness index ($r = .02$, $p = .514$)", so a drop in eeriness can be read independently of a drop in perceived human likeness.

✓ Read at primary-source depth during the 2026-08-21 audit.

17 gonzalez2018

Gonzalez, Rafael C., Woods, Richard E.. *Digital Image Processing*. Pearson. 2018.

DOI / URL	ISBN 9780133356724
Local copy	`05_image_processing/Gonzales,Woods-Digital.Image.Processing.4th.Edition.pdf`
Cited in	Ch.2 Background → Edge Detection Ch.3 Related Work → NPR for Virtual Humans Ch.4 Methodology → NPR Technique Design (3 citations)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

18 roberts1963

Roberts, L. G.. *Machine Perception of Three-Dimensional Solids*. 1963.

DOI / URL	*none recorded*
Local copy	`05_image_processing/33959125-MIT.pdf`
Cited in	Ch.2 Background → Edge Detection Ch.4 Methodology → NPR Technique Design (2 citations)

What it is. MIT doctoral thesis, "Machine Perception of Three-Dimensional Solids". Source of the Roberts Cross operator.

What it establishes. The operator as originally defined applies a square-root intensity transform first, then computes the two diagonal differences across a 2×2 neighbourhood. Read as an 82-page image scan; the operator definition is on p. 26.

✓ Read at primary-source depth during the 2026-08-21 audit.

19 itu709

International Telecommunication Union, Radiocommunication Sector. *Recommendation ITU-R BT.709-6: Parameter Values for the HDTV Standards for Production and International Programme Exchange*. 2015.

DOI / URL *none recorded*
Local copy *Online source, no PDF held.*
Cited in Ch.2 Background → Edge Detection (1 citation)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

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20 marr1980

Marr, David, Hildreth, Ellen. *Theory of Edge Detection*. Proceedings of the Royal Society of London. Series B, Biological Sciences. vol. 207, no. 1167, pp. 187–217. 1980.

DOI / URL `https://doi.org/10.1098/rspb.1980.0020`
Local copy `MarrHildreth (2).pdf`
Cited in Ch.2 Background → Edge Detection (1 citation)

What it is. Theory of edge detection built on detecting intensity changes at multiple scales, with edges located as zero-crossings of a Laplacian-of-Gaussian operator.

What it establishes. The rationale for the Gaussian is **not noise** — the word barely appears in the argument. It is the uncertainty principle: "these two localization requirements, the one in the spatial and the other in the frequency domain, are conflicting. They are, in fact, related by the uncertainty principle... There is, moreover, only one distribution that optimizes this relation, namely the Gaussian" (p. 192). Their operator is $\nabla^2 G$ with zero-crossing detection, not a gradient operator.

✓ Read at primary-source depth during the 2026-08-21 audit.

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21 canny1986

Canny, John. *A Computational Approach to Edge Detection*. IEEE Transactions on Pattern Analysis and Machine Intelligence. vol. PAMI-8, no. 6, pp. 679–698. 1986.

DOI / URL `https://doi.org/10.1109/TPAMI.1986.4767851`
Local copy `05_image_processing/A_Computational_Approach_to_Edge_Detection-sz4.pdf`
Cited in Ch.2 Background → Edge Detection (1 citation)

What it is. Derives an optimal step-edge operator from three criteria (good detection, good localisation, single response), shows its impulse response approximates the first derivative of a Gaussian, and builds a complete detector around it.

What it establishes. The pipeline does not end at non-maximum suppression. "In the current algorithm, no attempt is made to presegment contours. Instead the thresholding is done **with hysteresis**. If any part of a contour is above a high threshold, those points are immediately output, as is the entire connected segment of contour which contains the points and which lies above a low threshold." The full detector also combines multiple operator widths through feature synthesis.

✓ Read at primary-source depth during the 2026-08-21 audit.

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22 kyprianidis2009

Kyprianidis, Jan Eric, Kang, Henry, Döllner, Jürgen. *Image and Video Abstraction by Anisotropic Kuwahara Filtering*. Computer Graphics Forum. vol. 28, no. 7, pp. 1955–1963. 2009.

DOI / URL `https://doi.org/10.1111/j.1467-8659.2009.01574.x`
Local copy `03_npr_rendering/anisotropic_kuwahara.pdf`
Cited in Ch.2 Background → Structure-Preserving Filtering Ch.3 Related Work → NPR for Virtual Humans Ch.4 Methodology → NPR Technique Design Ch.5 Implementation → NPR Technique Implementation (4 citations)

What it is. Generalises the Kuwahara filter by adapting its kernel to local feature shape derived from the smoothed structure tensor, giving painterly abstraction that flows along image structure. Real-time GPU implementation, temporally coherent for video.

What it establishes. States the property that motivates using Kuwahara at all (p. 2): "the Kuwahara filter properly removes detail in **high-contrast** regions while also protecting shape boundaries in **low-contrast** regions. Therefore, it helps to maintain a roughly uniform level of abstraction across the image." This is the opposite of the bilateral filter, which "smooth[s] low-contrast regions while preserving high-contrast edges". Method: structure tensor gives orientation and anisotropy; the sampling window becomes an ellipse rotated to the local feature direction, $N = 8$ sectors, eccentricity scaled by anisotropy.

✓ Read at primary-source depth during the 2026-08-21 audit.

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23 kuwahara1976

Kuwahara, M., Hachimura, K., Eiho, S., Kinoshita, M.. *Processing of RI-Angiocardiographic Images*. In: Digital Processing of Biomedical Images. pp. 187–203. Plenum Press. 1976.

DOI / URL	`https://doi.org/10.1007/978-1-4684-0769-3_13`
Local copy	`03_npr_rendering/anisotropic_kuwahara.pdf`
Cited in	Ch.2 Background → Structure-Preserving Filtering (1 citation)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

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24 kyprianidis2011

Kyprianidis, J. E.. *Image and Video Abstraction by Multi-Scale Anisotropic Kuwahara Filtering*. In: Proceedings of the 9th International Symposium on Non-Photorealistic Animation and Rendering. pp. 55–64. ACM. 2011.

DOI / URL	ISBN 978-1-4503-0907-3
Local copy	`03_npr_rendering/anisotropic_multicale_209.pdf`
Cited in	Ch.2 Background → Structure-Preserving Filtering Ch.3 Related Work → NPR for Virtual Humans (2 citations)

What it is. Extends the anisotropic Kuwahara filter to multiple scales using an image pyramid processed coarse-to-fine.

What it establishes. Contains the precise description of the classical filter that the Background chapter relies on: "The general idea behind this filter is to divide the filter kernel into **four rectangular subregions that overlap by one pixel**. The filter response is then defined by the mean of a subregion with minimum variance." It credits [Kuwahara et al. 1976] in the same sentence, which matters because no copy of that 1976 chapter is held.

✓ Read at primary-source depth during the 2026-08-21 audit.

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25 macdorman2006

MacDorman, Karl F., Ishiguro, Hiroshi. *The Uncanny Advantage of Using Androids in Cognitive and Social Science Research*. Interaction Studies. vol. 7, no. 3, pp. 297–337. John Benjamins. 2006.

DOI / URL	*none recorded*
Local copy	`01_uncanny_valley/MacDorman2006AndroidScience.pdf`
Cited in	Ch.3 Related Work → Research Gap Ch.3 Related Work → The Uncanny Valley and Avatar Realism Ch.3 Related Work → Trust and Credibility in Virtual Agents (3 citations)

What it is. Position paper proposing the field of "android science", with one experiment attached. Argues an android is an experimental apparatus "controlled more precisely than any human actor" (p. 2), a perfect actor permitting scientists to vary parameters precisely, with the added advantage of physical presence.

What it establishes. The experiment is the first empirical reproduction of Mori's curve. Because robots spanning the likeness range did not exist (p. 8), they built the continuum photographically: two sequences of 11 morphed images (Qrio → Philip K. Dick android → the author; Eveliee → Repliee Q1Expo → its human model). 45 participants rated every image on nine-point human likeness and familiarity scales, then marked which felt eerie and rated those on a ten-point scale. Familiarity fell into a valley in both sequences with

the trough between humanoid and android, human likeness rose monotonically, and eeriness peaked where familiarity troughed.

✓ Read at primary-source depth during the 2026-08-21 audit.

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26 mcdonnell2009

McDonnell, R., Jörg, S., McHugh, J., Newell, F. N., O'Sullivan, C.. *Investigating the Role of Body Shape on the Perception of Emotion*. ACM Transactions on Applied Perception. vol. 6, no. 3, art. 14. 2009.

DOI / URL	`https://doi.org/10.1145/1577755.1577757`
Local copy	`01_uncanny_valley/McDonnell09_HappyZombies.pdf`
Cited in	Ch.3 Related Work → The Uncanny Valley and Avatar Realism (1 citation)

What it is. Real and virtual replicas of one actor performing six emotions, applied to five virtual bodies (low- and high-resolution counterpart, cartoon, wooden mannequin, zombie) plus the original video, and a point-light figure with no form.

What it establishes. Headline finding, abstract p. 1: emotional action perception is "highly robust and to the most part independent of the character's body, **so long as form is present**." The point-light condition is the exception: main effects of body for happiness, disgust and surprise with lower rated intensity (p. 8), and emotions "recognized correctly more often on the real and virtual bodies than points".

✓ Read at primary-source depth during the 2026-08-21 audit.

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27 dill2012

Dill, V., Flach, L. M., Hocevar, R., Lykawka, C., Musse, S. R., Pinho, M. S.. *Evaluation of the Uncanny Valley in CG Characters*. In: Intelligent Virtual Agents: 12th International Conference, IVA 2012. vol. 7502, pp. 511–513. Springer. 2012.

DOI / URL	*none recorded*
Local copy	`01_uncanny_valley/Evaluation of the Uncanny Valley in CG Characters.pdf`
Cited in	Ch.3 Related Work → The Uncanny Valley and Avatar Realism Ch.6 Evaluation → Discussion (2 citations)

What it is. Validates the uncanny valley for CG characters drawn from contemporary commercial films and games. Familiarity ratings followed Mori's predicted curve with the strongest negative response in the near-human range.

What it establishes. Motion deepened the valley most visibly for two of the ten characters tested, not for every stimulus. Participants named the eyes and the mouth as the facial regions producing the most discomfort.

✓ Read at primary-source depth during the 2026-08-21 audit.

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28 seymour2021

Seymour, Mike, Yuan, Lingyao, Dennis, Alan R., Riemer, Kai. *Have We Crossed the Uncanny Valley? Understanding Affinity, Trustworthiness, and Preference for Realistic Digital Humans in Immersive Environments*. Journal of the Association for Information Systems. vol. 22, no. 3, pp. 591–617. 2021.

DOI / URL	`https://doi.org/10.17705/1jais.00491`
Local copy	`01_uncanny_valley/Seymour, Yuan, Dennis & Riemer, "Have We Crossed the Uncanny Valley?" (2021).pdf`
Cited in	Ch.3 Related Work → The Uncanny Valley and Avatar Realism (1 citation)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

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29 kuffner2024

Kuffner dos Anjos, R., Madeiras Pereira, J.. *Effects of Realism and Representation on Self-Embodied Avatars in Immersive Virtual Environments*. 2024.

DOI / URL *none recorded*
Local copy `02_trust_and_avatar_perception/Effects of Realism and Representation on Self-Embodied Avatars in Immersive Virtual Environments.pdf`
Cited in Ch.3 Related Work → The Uncanny Valley and Avatar Realism (1 citation)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

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30 tao2025

Tao, Z., Liu, Y., Qiu, J., Li, S.. *Impact of Virtual Avatar Appearance Realism on Perceptual Interaction Experience: A Network Meta-Analysis*. Frontiers in Psychology. vol. 16, pp. 1624975. 2025.

DOI / URL `https://doi.org/10.3389/fpsyg.2025.1624975`
Local copy `Tao-Impact.pdf`
Cited in Ch.3 Related Work → The Uncanny Valley and Avatar Realism (1 citation)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

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31 alipour2026

Alipour, Ahdiyeh, Hartmann, Tilo, Alimardani, Maryam. *Would You Rely on an Eerie Agent? A Systematic Review of the Impact of the Uncanny Valley Effect on Trust in Human–Agent Interaction*. Computers in Human Behavior Reports, vol. 23, art. 101220, 2026.

DOI / URL DOI `10.1016/j.chbr.2026.101220` (preprint: `https://arxiv.org/abs/2505.05543`)
Local copy `02_trust_and_avatar_perception/Alipor_Would You Rely on an Eerie Agent?.pdf`
Cited in Ch.3 Related Work → Research Gap Ch.3 Related Work → Trust and Credibility in Virtual Agents Ch.4 Methodology → User Study Design Ch.6 Evaluation → User Study Results (4 citations)

What it is. PRISMA systematic review of 53 empirical studies asking whether the uncanny valley effect influences trust in agents.

What it establishes. Codes studies along several independent dimensions, each with its own count. **Trust measurement: three groups** — direct measures ($n = 24$), trust-related constructs such as competence and credibility ($n = 11$), trust-related intentions such as advice adherence ($n = 18$). **Modality: four types** — visual, auditory, tactile, textual. **Effect type: three** — of the 41 studies that tested it empirically, 17 reported a direct effect, 24 a conditional or indirect one (10 dual-coded), and 10 none. Only 7 of 53 used an objective behavioural measure alongside self-report.

✓ Read at primary-source depth during the 2026-08-21 audit.

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32 nowak2003

Nowak, Kristine L., Biocca, Frank. *The Effect of the Agency and Anthropomorphism on Users' Sense of Telepresence, Copresence, and Social Presence in Virtual Environments*. Presence: Teleoperators and Virtual Environments. vol. 12, no. 5, pp. 481–494. 2003.

DOI / URL `https://doi.org/10.1162/105474603322761289`
Local copy `02_trust_and_avatar_perception/Nowak2003.pdf`
Cited in Ch.3 Related Work → Trust and Credibility in Virtual Agents (1 citation)

What it is. 3×2 experiment. Anthropomorphism of a partner's image (high / low / none) crossed with per-

ceived agency (told the partner was a human-driven avatar or a computer-driven agent). Four outcomes: perceived other's copresence, self-reported copresence, social presence, telepresence.

What it establishes. Agency was null on all four ($p = .7$, $.57$, $.33$, $.46$); the discussion section is headed "Agency Does Not Influence Presence". Anthropomorphism ran *opposite* to their own prediction: Hypotheses V and VI were "not supported. . . significant in the opposite direction", with the less anthropomorphic image producing more copresence ($R = .22$, $p = .01$) and more social presence than the highly anthropomorphic one.

✓ Read at primary-source depth during the 2026-08-21 audit.

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33 alimardani2024

Alimardani, M., de Roode, R., Vaitonytė, J., Louwerse, M. M.. *Effect of a Virtual Agent's Appearance and Voice on Uncanny Valley and Trust in Human-Agent Collaboration*. In: Proceedings of the 24th ACM International Conference on Intelligent Virtual Agents. art. 5. ACM. 2024.

DOI / URL	`https://doi.org/10.1145/3652988.3673970`
Local copy	`02_trust_and_avatar_perception/Mardani Effect of a Virtual AgentâĂŹs Appearance and Voi pdf`
Cited in	Ch.3 Related Work → Trust and Credibility in Virtual Agents (1 citation)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

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34 gao2025

Gao, Yang, Dai, Yangbin, Zhang, Guangtao, Guo, Honglei, Mostajeran, Fariba, Zheng, Binge, Yu, Tao. *Trust in Virtual Agents: Exploring the Role of Stylization and Voice*. IEEE Transactions on Visualization and Computer Graphics. vol. 31, no. 5, pp. 3623–3633. 2025.

DOI / URL	`https://doi.org/10.1109/TVCG.2025.3549566`
Local copy	`02_trust_and_avatar_perception/IEEE_VR25_TVCG.pdf`
Cited in	Ch.3 Related Work → Trust and Credibility in Virtual Agents Ch.4 Methodology → User Study Design (2 citations)

What it is. 27 participants in VR watch a 40–60 s TED-style talk. Style (Real / Stylization / No-Stylization) and Voice within-subjects, Identity between-subjects. **Real is the original recorded video of the speaker**, Stylization is an agent reconstructed from that speaker's own images, No-Stylization a stock MetaHuman.

What it establishes. Real scored significantly higher on Ability ($p < 0.001$), Benevolence ($p = 0.007$) and Integrity ($p = 0.015$). Only the single global trust item failed to separate Real from Stylization ($p = 0.216$). The abstract's claim of "comparable" trust holds for that one item only. The discussion is explicit: "This contrasts with previous studies. . . Our results indicate that using a real-look agent can increase trust levels." Structurally this is the closest published analogue to the C1-versus-C3 comparison in this thesis.

✓ Read at primary-source depth during the 2026-08-21 audit.

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35 dubosc2025

Dubosc, C., Gorisse, G., Lefrou, T., Richir, S., Christmann, O.. *Effect of Avatar Stylization and Facial Expression Intensity in Virtual Interactions*. Virtual Reality. 2025.

DOI / URL	`https://doi.org/10.1007/s10055-025-01238-6`
Local copy	`02_trust_and_avatar_perception/Effect_of_avatar_stylization_and_facial_expression.pdf`
Cited in	Ch.3 Related Work → Trust and Credibility in Virtual Agents (1 citation)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

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36 cihodaru2025

Cihodaru-Ştefanache, Ş., Podina, I. R.. *The Uncanny Valley Effect in Embodied Conversational Agents: A Critical Systematic Review of Attractiveness, Anthropomorphism, and Uncanniness*. Frontiers in Psychology. vol. 16, pp. 1625984. 2025.

DOI / URL	`https://doi.org/10.3389/fpsyg.2025.1625984`
Local copy	`01_uncanny_valley/The_uncanny_valley_effect_in_embodied_conversation.pdf`
Cited in	Ch.3 Related Work → Trust and Credibility in Virtual Agents Ch.6 Evaluation → Discussion (2 citations)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

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37 kumar2019

Pavan Kumar, M. P., Poornima, B., Nagendraswamy, H. S., Manjunath, C.. *A Comprehensive Survey on Non-Photorealistic Rendering and Benchmark Developments for Image Abstraction and Stylization*. Iran Journal of Computer Science. vol. 2, pp. 131–165. 2019.

DOI / URL	`https://doi.org/10.1007/s42044-019-00034-1`
Local copy	`03_npr_rendering/Kumar- A comprehensive survey on NPR.pdf`
Cited in	Ch.3 Related Work → NPR for Virtual Humans (1 citation)

What it is. Comprehensive survey of NPR and benchmark development for image abstraction and stylisation, covering significant papers from 1963 to 2017. **Four authors**, not two.

What it establishes. Classifies NPR into four families: stroke-based rendering, colour image analogy, region-based rendering, and image filtering abstraction and stylisation. Coverage is image-space throughout, so geometry-based approaches such as inverted hull outlines fall outside it — a limitation the thesis notes explicitly.

✓ Read at primary-source depth during the 2026-08-21 audit.

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38 lake2000

Lake, Adam, Marshall, Carl, Harris, Mark, Blackstein, Marc. *Stylized Rendering Techniques for Scalable Real-Time 3D Animation*. In: Proceedings of the 1st International Symposium on Non-Photorealistic Animation and Rendering (NPAR). pp. 13–20. 2000.

DOI / URL	`https://doi.org/10.1145/340916.340918`
Local copy	`03_npr_rendering/lake et al Stylized rendering.pdf`
Cited in	Ch.3 Related Work → NPR for Virtual Humans Ch.4 Methodology → NPR Technique Design (2 citations)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

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39 barla2006

Barla, Pascal, Thollot, Joëlle, Markosian, Lee. *X-Toon: An Extended Toon Shader*. In: Proceedings of the 4th International Symposium on Non-Photorealistic Animation and Rendering (NPAR). pp. 127–132. 2006.

DOI / URL	`https://doi.org/10.1145/1124728.1124749`
Local copy	`03_npr_rendering/x-toon.pdf`
Cited in	Ch.3 Related Work → NPR for Virtual Humans Ch.4 Methodology → NPR Technique Design (2 citations)

What it is. Extends toon shading by replacing the 1D tone texture with a 2D texture whose second axis car-

ries “tone detail”. Second, independent extension: a modified normal field interpolating between original and abstracted-shape normals, governed by a global shape detail parameter.

What it establishes. Coins the terminology the thesis uses: “toon shading does not reflect the desired **level of abstraction (LOA)** of a surface. . . We use the term LOA instead of LOD to emphasize that our goal is visual abstraction, not increased rendering speed.” The first ramp axis remains the Lambertian $\mathbf{n} \cdot \mathbf{l}$ term; only the second is new. The attribute mapping mechanism is general — “any attribute can be chosen” — and they demonstrate two, both view-dependent: depth-based and orientation-based ($D = |\mathbf{n} \cdot \mathbf{v}|^r$).

✓ Read at primary-source depth during the 2026-08-21 audit.

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40 raskar1999

Raskar, Ramesh, Cohen, Michael. *Image Precision Silhouette Edges*. In: Proceedings of the 1999 Symposium on Interactive 3D Graphics. pp. 135–140. ACM. 1999.

DOI / URL	`https://doi.org/10.1145/300523.300539`
Local copy	`03_npr_rendering/Raskar -Image Precision Silhouette Edges.pdf`
Cited in	Ch.3 Related Work → NPR for Virtual Humans (1 citation)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

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41 roshaan2026

Lino Roshaan M.K.. *AHEAD: Adaptive Hierarchical Edge Detection for Real-Time Artistic Stylization*. 2026.

DOI / URL	`https://doi.org/10.21203/rs.3.rs-8849147/v1`
Local copy	`03_npr_rendering/Adaptive Hierarchical Edge Detection.pdf`
Cited in	Ch.3 Related Work → NPR for Virtual Humans Ch.4 Methodology → NPR Technique Design Ch.5 Implementation → NPR Technique Implementation (3 citations)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

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42 wisessing2016

Wisessing, P., Dingliana, J., McDonnell, R.. *Perception of Lighting and Shading for Animated Virtual Characters*. In: Proceedings of the ACM Symposium on Applied Perception. pp. 25–29. ACM. 2016.

DOI / URL	`https://doi.org/10.1145/2931002.2931015`
Local copy	`Wissesing- perception.pdf`
Cited in	Ch.3 Related Work → NPR for Virtual Humans (1 citation)

What it is. Two shading styles (CG smooth-shaded vs toon-shaded) crossed with five lighting conditions, on a single greyscale cartoon character (Mery). **No photorealistic condition exists in this study.**

What it establishes. Main finding, p. 4: “the shading style used (CG vs. toon) does not change the recognition of emotion, but does change the perception of intensity of that emotion, with toon-shaded renders being rated as displaying emotions as less intense than CG-shaded across the board.” Recognition accuracy was unaffected by shading style or lighting.

✓ Read at primary-source depth during the 2026-08-21 audit.

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43 wisessing2020

Wisessing, P., Zibrek, K., Cunningham, D. W., Dingliana, J., McDonnell, R.. *Enlighten Me: Importance of Brightness and Shadow for Character Emotion and Appeal*. ACM Transactions on Graphics. vol. 39, no. 3, art. 19, pp. 19:1–19:12. 2020.

DOI / URL `https://doi.org/10.1145/3383195`
Local copy `Wissessing-Enloghten me.pdf`
Cited in Ch.3 Related Work → NPR for Virtual Humans (1 citation)

What it is. Extensive perceptual experiments on key-light brightness and key-to-fill ratio for emotional CG characters, including a follow-up set with realistic characters.

What it establishes. Findings bullets: "Higher brightness makes for more appealing characters, regardless of the emotion they portray"; brightness alters perceived intensity in opposite directions for happy and sad; "Darker shadows are unappealing for cartoon characters but do not affect the appeal of realistic characters". Darker key-to-fill ratios also raised eeriness ($16{:}1$ more eerie than $4{:}1$ or $1{:}1$).

✓ Read at primary-source depth during the 2026-08-21 audit.

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44 zibrek2019tap

Zibrek, Katja, Martin, Sean, McDonnell, Rachel. *Is Photorealism Important for Perception of Expressive Virtual Humans in Virtual Reality?*. ACM Transactions on Applied Perception. vol. 16, no. 3, pp. 14:1–14:19. 2019.

DOI / URL `https://doi.org/10.1145/3349609`
Local copy `02_trust_and_avatar_perception/TAP_2019.pdf`
Cited in Ch.3 Related Work → NPR for Virtual Humans (1 citation)

What it is. Large **between-group** VR study: $N = 797$ recruited, 622 analysed, run in a public science gallery. Each participant saw *one* of three render styles (Realistic, Simple, Sketch) in one of three scenarios (Friendly, Unfriendly, Sad) — a 3×3 design chosen deliberately "to reduce learning and transfer across the experimental conditions".

What it establishes. Thirteen measured scales, none of which is emotion *recognition*; the emotional items measure the participant's own affective state ("The girl I just observed made me feel concerned"). Social presence: "we found no main effects or interactions between render style or scenario". Affinity (appeal, familiarity, eeriness): "We did not find any effects of render style". The headline is that photorealism *changed* emotional response rather than improving it, and the abstract reports "a preference for realism in VR".

✓ Read at primary-source depth during the 2026-08-21 audit.

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45 zibrek2019mig

Zibrek, Katja, McDonnell, Rachel. *Social Presence and Place Illusion are Affected by Photorealism in Embodied VR*. In: Proceedings of the 12th ACM SIGGRAPH Conference on Motion, Interaction and Games (MIG). pp. 1–7. ACM. 2019.

DOI / URL `https://doi.org/10.1145/3359566.3360064`
Local copy `02_trust_and_avatar_perception/3359566.3360064.pdf`
Cited in Ch.3 Related Work → NPR for Virtual Humans Ch.6 Evaluation → Discussion (2 citations)

What it is. Within-subjects VR study in which participants are embodied in a high-fidelity virtual body and observe an animated character rendered photorealistically or in a simplified style, with eye-gaze interaction.

What it establishes. "Our goal was to evaluate if photorealism alone is enough to increase self-reported and behavioural signs (interpersonal distance or proximity) of social presence, place illusion, and empathetic concern... This was found to be the case for self-reported social presence and place illusion." The behavioural proximity measure was unaffected. The contrast with the TAP study is embodiment: social presence responds to render style only once the viewer inhabits the space.

✓ Read at primary-source depth during the 2026-08-21 audit.

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46 latoschik2017

Latoschik, Marc Erich, Roth, Daniel, Gall, Dominik, Achenbach, Jascha, Waltemate, Thomas, Botsch, Mario. *The Effect of Avatar Realism in Immersive Social Virtual Realities*. In: Proceedings of the 23rd ACM Symposium on Virtual Reality Software and Technology (VRST). ACM. 2017.

DOI / URL `https://doi.org/10.1145/3139131.3139156`
Local copy `02_trust_and_avatar_perception/2017-vrst-effect-of-avatar-realism.pdf`
Cited in Ch.3 Related Work → NPR for Virtual Humans Ch.4 Methodology → User Study Design (2 citations)

What it is. VRST study on avatar realism in immersive social VR, using a 2×2 within-subjects design crossing self-avatar with other-avatar realism.

What it establishes. Design confirmed from the paper: "a 2 x (self-avatar) x 2 (other-avatar) within-subjects design", analysed with "two factorial ANOVAs for repeated measures on all measures taken". Cited in this thesis as precedent for the repeated-measures ANOVA on within-participant rendering-style comparisons.

✓ Read at primary-source depth during the 2026-08-21 audit.

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47 zell2015

Zell, Eduard, Aliaga, Carlos, Jarabo, Adrian, Zibrek, Katja, Gutierrez, Diego, McDonnell, Rachel, Botsch, Mario. *To Stylize or Not to Stylize? The Effect of Shape and Material Stylization on the Perception of Computer-Generated Faces*. ACM Transactions on Graphics. vol. 34, no. 6, pp. 184:1–184:12. ACM. 2015.

DOI / URL `https://doi.org/10.1145/2816795.2818126`
Local copy `01_uncanny_valley/To Stylize or not to Stylize?_Zell.pdf`
Cited in Ch.3 Related Work → NPR for Virtual Humans Ch.4 Methodology → User Study Design Ch.6 Evaluation → Discussion (3 citations)

What it is. Separates a character's shape from its material and varies each independently. First round used three stylisation levels, the extended round five, across two characters and five expressions.

What it establishes. Shape governs perceived realism; material governs appeal and eeriness. Strong mismatches between the two levels produce very unappealing characters, with realistic material on realistic shape the least appealing combination and cartoon material on realistic shape similarly poor. On expressions: "The perceived intensity of expressions *decreases with realism of shape*" — so cartoon shapes are rated *more* intense — and all expressions except neutral were "recognized well or outstandingly well independent of the shape".

✓ Read at primary-source depth during the 2026-08-21 audit.

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48 mixamo

Adobe Inc.. *Mixamo: 3D Character Rigging and Animation Service*. 2024.

DOI / URL *none recorded*
Local copy `https://www.mixamo.com/`
Cited in Ch.4 Methodology → Avatar Platforms and Rendering Constraints (1 citation)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

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49 metasdk

Meta Platforms, Inc.. *Meta Avatars SDK Documentation*. 2024.

DOI / URL *none recorded*
Local copy `https://developer.oculus.com/documentation/unity/meta-avatars-overview/`
Cited in Ch.4 Methodology → Avatar Platforms and Rendering Constraints (1 citation)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

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50 avaturn

Avaturn. *What is Avaturn?*. 2024.

DOI / URL *none recorded*
Local copy `https://docs.avaturn.me/docs/what-is-avaturn/`
Cited in Ch.4 Methodology → Avatar Platforms and Rendering Constraints (1 citation)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

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51 gltf2

Khronos Group. *glTF 2.0 Specification*. 2017.

DOI / URL *none recorded*
Local copy *Online source, no PDF held.*
Cited in Ch.4 Methodology → Avatar Platforms and Rendering Constraints (1 citation)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

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52 saitotakahashi1990

Saito, Takafumi, Takahashi, Tokiichiro. *Comprehensible Rendering of 3-D Shapes*. In: Proceedings of the 17th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH). pp. 197–206. ACM. 1990.

DOI / URL `https://doi.org/10.1145/97880.97901`
Local copy `03_npr_rendering/Comprehensible Rendering of 3-D Shapes Saito.pdf`
Cited in Ch.4 Methodology → NPR Technique Design (1 citation)

What it is. Introduces the **G-buffer**: instead of shading directly to colour, geometric properties are rendered into 2D buffers (normals n_x, n_y, n_z, depth s_z, surface parameters, object/patch identifiers), then ordinary 2D image-processing operators run over those buffers and the results are composited.

What it establishes. Where the edges come from matters: “Profiles and edges are the 0-th and first order discontinuities of the **depth image (s_z image)**”. The alternative route is the object/patch identifier image. In their worked example “The shaded image (sh) was calculated from the n_x, n_y and n_z images. The profile image (pr) was obtained from the s_z image” — so the normal channels drive *shading*, and no differential operator is applied to them.

✓ Read at primary-source depth during the 2026-08-21 audit.

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53 nienhaus2003

Nienhaus, Marc, Döllner, Jürgen. *Edge-Enhancement: An Algorithm for Real-Time Non-Photorealistic Rendering*. Journal of WSCG. vol. 11, no. 1, pp. 346–353. 2003.

DOI / URL *none recorded*
Local copy `03_npr_rendering/Edge Enhancement.pdf`
Cited in Ch.4 Methodology → NPR Technique Design (1 citation)

What it is. Builds an *edge map*, a 2D texture encoding silhouette, border and crease edges, extracted from G-buffer-like data using texture hardware so it runs in real time, then applied to scene objects by projective texturing.

What it establishes. Splits the cues explicitly (p. 3): “abrupt changes in the z-buffer occur at silhouette edges and border edges, i.e., 0th order discontinuities of the z-buffer indicate profiles. . . **Abrupt changes in the normal-buffer occur typically at crease edges**. . . 0th order discontinuities of the normal-buffer indicate

crease edges." Credits Saito and Takahashi for the G-buffer.

✓ Read at primary-source depth during the 2026-08-21 audit.

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54 `winnemoller2012`

Winnemöller, Holger, Kyprianidis, Jan Eric, Olsen, Sven C.. *XDoG: An eXtended Difference-of-Gaussians Compendium Including Advanced Image Stylization*. Computers & Graphics. vol. 36, no. 6, pp. 740–753. 2012.

DOI / URL	`https://doi.org/10.1016/j.cag.2012.03.004`
Local copy	`Winnemoeller12.pdf`
Cited in	Ch.4 Methodology → NPR Technique Design Ch.5 Implementation → NPR Technique Implementation (2 citations)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

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55 `demsar2006`

Demšar, Janez. *Statistical Comparisons of Classifiers over Multiple Data Sets*. Journal of Machine Learning Research. vol. 7, pp. 1–30. 2006.

DOI / URL	`https://www.jmlr.org/papers/v7/demsar06a.html`
Local copy	`06_statistics/demsar06a.pdf`
Cited in	Ch.4 Methodology → Parameter Adjustment Study Design Ch.4 Methodology → User Study (2 citations)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

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56 `lakens2017`

Lakens, Daniël. *Equivalence Tests: A Practical Primer for t Tests, Correlations, and Meta-Analyses*. Social Psychological and Personality Science. vol. 8, no. 4, pp. 355–362. 2017.

DOI / URL	`https://doi.org/10.1177/1948550617697177`
Local copy	`Lakens -equivalence Test.pdf`
Cited in	Ch.4 Methodology → User Study Design (1 citation)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.

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57 `blinn1977`

Blinn, James F.. *Models of Light Reflection for Computer Synthesized Pictures*. In: Proceedings of the 4th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH). pp. 192–198. 1977.

DOI / URL	`https://doi.org/10.1145/965141.563893`
Local copy	`Blinn.pdf`
Cited in	Ch.5 Implementation → NPR Technique Implementation (1 citation)

Bibliographic record only. This source has not been re-read at primary-source depth, so no summary is asserted here. Its per-sentence verification lives in `citation_check.pdf`.

○ Bibliographic fields verified; content summary not yet written.
